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Final Project

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Abstract

This project involves designing and configuring a computer network topology in Cisco Packet Tracer that integrates multiple routers, switches, PCs, and a server to meet specific connectivity, VLAN segmentation, IP addressing, and routing requirements. The design focuses on applying key networking concepts such as VLAN configuration, DHCP services, OSPF routing, NAT, PAT, and Access Control Lists (ACLs) to create a secure and functional network. The final implementation ensures seamless communication between authorized devices while restricting access where required, thereby simulating areal-world enterprise networkenvironment.

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Introduction

Computer networks form the backbone of modern communication, connecting devices, systems, and users across local and global distances. In an enterprise environment, network design must not only provide seamless connectivity but also ensure security, scalability, and efficient resource utilization.
This project focuses on designing and configuring a network topology in Cisco Packet Tracer that incorporates routers, switches, PCs, and a server, simulating a realistic business scenario. The network implements VLAN segmentation, DHCP for automated IP address allocation, OSPF routing for dynamic path determination, and NAT/PAT for Internet access. Access Control Lists (ACLs) are configured to regulate traffic flow and enforce security policies. Through this project, students gain hands-on experience in applying theoretical networking concepts to practical scenarios, enhancing both technical knowledge and problem-solving skills.

Objectives

- To design and implement a multi-router and multi-switch topology using Cisco Packet Tracer.
- To configure VLANs, IP addressing, and trunking for logical network segmentation.
- To apply dynamic and static IP assignment using DHCP and manual configuration.
- To implement OSPF routing for network-wide communication.
- To configure NAT, PAT, and ACLs for network security and controlled access.
- To ensure remote management of network devices via Telnet/SSH.

Theory

In enterprise networking, logical segmentation and secure communication are essential to maintain performance and security. VLANs divide a network into smaller broadcast domains, enhancing efficiency and security. DHCP automates IP address allocation, reducing administrative overhead. Routing protocols like OSPF allow dynamic route exchange, ensuring adaptability and redundancy. NAT and PAT enable private networks to access external resources while conserving public IPs. ACLs are used to control traffic and enforce security policies. This project applies these concepts in a simulated environment, reflecting best practices in network design.

Procedure and Discussions

Creating and Assigning VLANs

VLANs 110 and 120 on Switch0

```
Switch(config)#vlan 110
Switch(config-vlan)#name Engineers
Switch(config-vlan)#vlan 120
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
```

```
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 110
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 110
Switch(config-if)#end
```

```
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 120
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 120
Switch(config-if)#
Switch(config-if)#end
```

Show VLANs created

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
110  Engineers                        active    Fa0/1, Fa0/2
120  IT                               active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

VLAN 130 on Switch1 and showing result

```
Switch(config)# vlan 130
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 130
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 130
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
130  HR                               active    Fa0/1, Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Creating VLAN 4 and 30 on Switch2

```
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
Switch(config)#int fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#exit
```

```
Switch(config)#vlan 30
Switch(config-vlan)#name VLAN30
Switch(config-vlan)#exit
Switch(config)#int fa0/24
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
```

Showing result

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Gig0/1, Gig0/2
4    VLAN4                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5
30   VLAN30                           active    Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

IP Configurations

IPsforPCs 4-10wereconfigured manually by going to PC# > Desktop > IP configuration

Router0 IP Configuration:

We used encapsulation command to let this subinterface know that it will handle traffic from the specified VLAN.

```
Router(config)#int se0/1/0
Router(config-if)#ip address 200.0.0.1 255.255.255.252
Router(config-if)#exit
Router(config)#int g0/0/0.110
Router(config-subif)#encapsulation dot1Q 110
Router(config-subif)#ip address 110.0.0.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int g0/0/0.120
Router(config-subif)#encapsulation dot1Q 120
Router(config-subif)#ip address 120.0.0.1 255.255.255.0
Router(config-subif)#end
```

Result:

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
GigabitEthernet0/0/0.110110.0.0.1       YES manual administratively down down
GigabitEthernet0/0/0.120120.0.0.1       YES manual administratively down down
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            200.0.0.1       YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Router1 IP Configuration:

```
Router(config)#int se0/1/0
Router(config-if)#no ip address 192.168.1.2 255.255.255.252
Router(config-if)#ip address 200.0.0.2 255.255.255.252
Router(config-if)#exit
Router(config)#int g0/0/0
Router(config-if)#ip address 30.0.0.1 255.255.255.0
Router(config-if)#exit
```

```
Router(config)#int g0/0/0.130
Router(config-subif)#encapsulation dot1Q 130
Router(config-subif)#ip address 130.0.0.1 255.255.255.0
Router(config-subif)#end
```

Result

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   30.0.0.1        YES manual administratively down down
GigabitEthernet0/0/0.130130.0.0.1       YES manual administratively down down
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            200.0.0.2       YES manual up                    up
Serial0/1/1            201.0.0.1       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router2 IP configuration

```
Router(config)#int se0/1/1
Router(config-if)#ip address 201.0.0.2 255.255.255.252
Router(config-if)#no shut
```

```
Router(config)#int g0/0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int g0/0/0.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#end
```

Result

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
GigabitEthernet0/0/0.4 192.168.4.1     YES manual administratively down down
GigabitEthernet0/0/0.30192.168.30.1    YES manual administratively down down
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES manual down                  down
Serial0/1/1            201.0.0.2       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Switch interface for VLAN 1

```
Switch(config)#interface vlan 1
Switch(config-if)#ip address 30.0.0.5 255.255.255.0
Switch(config-if)#end
```

Result

```
Switch#show ip interface vlan 1
Vlan1 is administratively down, line protocol is down
  Internet address is 30.0.0.5/24
```

***All interfaces are later set to no shutdown**

Trunking

Procedure is shown for switch 1 and router 1. Same procedure is done for switches 0 and 2 and routers 0 and 2.

Switch 1:

```
Switch(config)#int g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 130
Switch(config-if)#
```

Router 1:

```
Router(config)#int g0/0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.130, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.130, changed state to up
```

Result on all switches:

```
S0-Psut#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig0/1      110,120

Port        Vlans allowed and active in management domain
Gig0/1      110,120

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      110,120
```

```
S1-Psut#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      130
```

```
S2-Psut#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      4,30
```

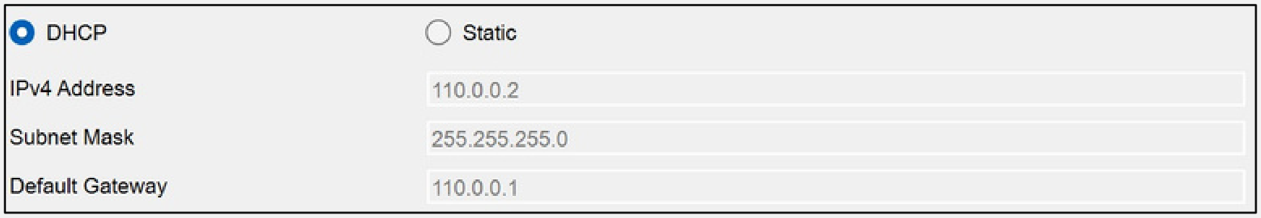
Configuring Router1 as DHCP server

```
Router(config)#ip dhcp excluded-address 110.0.0.1 110.0.0.1
Router(config)#ip dhcp pool POOL110
Router(dhcp-config)#network ip-ad
Router(dhcp-config)#network ip-add
Router(dhcp-config)#network 110.0.0.0 255.255.255.0
Router(dhcp-config)#default-router 110.0.0.1
Router(dhcp-config)#
Router(dhcp-config)#exit
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 120.0.0.1 120.0.0.1
Router(config)#ip dhcp pool POOL120
Router(dhcp-config)#network 120.0.0.0 255.255.255.0
Router(dhcp-config)#defau
Router(dhcp-config)#default-router 120.0.0.1
Router(dhcp-config)#
```

Result:

```
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
110.0.0.2        0090.2BDD.D497           --                     Automatic
110.0.0.3        0030.F2C2.DA39           --                     Automatic
120.0.0.2        0030.A37A.27BC           --                     Automatic
120.0.0.3        0060.47C2.DE04           --                     Automatic
Router#
```

IP configuration result on PC0:



Configure OSPF on Routers:

On**Router0:** router ospf 1

network 110.0.0.0 0.0.0.255 area 0

network 120.0.0.0 0.0.0.255 area 0

network 200.0.0.0 0.0.0.255 area 0

exit

On **Router1:** router ospf 1

network 200.0.0.0 0.0.0.255 area 0

network 201.0.0.0 0.0.0.255 area 0

network 30.0.0.0 0.0.0.255 area 0

network 130.0.0.0 0.0.0.255 area 0

exit

On **Router2:** router ospf 1

network 200.0.0.0 0.0.0.255 area 0

network 201.0.0.0 0.0.0.255 area 0

exit

Configure Hostnames

```
On Router0: hostname R0-Psut
On Router1: hostname R1-Psut
On Router2: hostname R2-Psut
On Switch0: hostname S0-Psut
On Switch1: hostname S1-Psut
On Switch2: hostname S2-Psut
```

Configure passwords

(enable,console,VTY)password is ‘PSUT’ on all routers and switches:

```
Config terminal
enable secret PSUT
line console 0
password PSUT
exit
line vty 0 4
password PSUT
exit
```

Remote access to Switch1 (S1-PSUT):

Aswitch shouldbe givenanIP address anddefault gateway on VLAN1 aka management interface to be accessed remotely. IP address is given previously.

On Switch1:

```
Config terminal
interface vlan 1
ip default-gateway 30.0.0.1
```

Static NAT on Router2 for Server

Server (192.168.30.10)isinaprivatenetwork and uses public IP on S0/1/1 for communication.

```
Config terminal
ip nat inside source static 192.168.30.10 201.0.0.2
interface GigabitEthernet0/0/0.30
ip nat inside
exit
interface Serial0/1/1
ip nat outside
exit
```

Check Static NAT configuration on Router2:

Using show running-config/show ip nat translations

```
ip nat inside source static 192.168.30.10 201.0.0.2
```

```
R2-Psut#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  201.0.0.2         192.168.30.10      ---                ---
```

To verify our NAT configuration, we tried to ping the server's IP address from a PC out of its network resulting in an ICMP reply from the public address of the server indicating NAT configuration worked.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 201.0.0.2: bytes=32 time=1ms TTL=126
Reply from 201.0.0.2: bytes=32 time=1ms TTL=126
Reply from 201.0.0.2: bytes=32 time=20ms TTL=126
Reply from 201.0.0.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 6ms
```

Ping the public IP from a PC4 outside the server network:

This tests whether the static NAT allows reaching the server via its public IP.

```
C:\>ping 172.40.0.6

Pinging 172.40.0.6 with 32 bytes of data:

Reply from 130.0.0.1: Destination host unreachable.
Reply from 130.0.0.1: Destination host unreachable.
Reply from 130.0.0.1: Destination host unreachable.
Reply from 130.0.0.1: Destination host unreachable.

Ping statistics for 172.40.0.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

The “Destination Host Unreachable” message that appears when pinging the server’s public IP from PC4 is expected in this scenario. This is because my static NAT configuration uses a different public IP mapping. Therefore, the unreachable result does not indicate an error, it reflects that the test IP does not match my actual static NAT configuration. The server remains accessible through its correct public IP as defined in my NAT settings.

PAT for PCs 6-10 using public IP 172.40.0.5:

```
Config terminal
access-list 1 permit 192.168.4.0 0.0.0.31
interface GigabitEthernet0/0/0.4
ip nat inside
exit
interface Serial0/1/1
ip nat outside
exit
ip nat inside source list 1 interface Serial0/1/1 overload
```

To verify PAT for PCs 6-10 sharing public IP 172.40.0.5:

a- Check the PAT configuration:

```
R2-Psut#show running-config
```

```
!
ip nat inside source list 1 interface Serial0/1/1 overload
```

```
interface GigabitEthernet0/0/0.4
 encapsulation dot1Q 4
 ip address 192.168.4.1 255.255.255.0
 ip nat inside
```

```
interface Serial0/1/1
 ip address 201.0.0.2 255.255.255.252
 ip nat outside
```

b- Generate traffic from each PC (we tried on PC6):

PC6

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.0.0.1

Pinging 200.0.0.1 with 32 bytes of data:

Reply from 200.0.0.1: bytes=32 time=2ms TTL=253
Reply from 200.0.0.1: bytes=32 time=2ms TTL=253
Reply from 200.0.0.1: bytes=32 time=7ms TTL=253
Reply from 200.0.0.1: bytes=32 time=4ms TTL=253

Ping statistics for 200.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms
```

c- Check NAT translations:

```
R2-Psut#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 201.0.0.2:1       192.168.4.2:1      200.0.0.1:1        200.0.0.1:1
icmp 201.0.0.2:2       192.168.4.2:2      200.0.0.1:2        200.0.0.1:2
icmp 201.0.0.2:3       192.168.4.2:3      200.0.0.1:3        200.0.0.1:3
icmp 201.0.0.2:4       192.168.4.2:4      200.0.0.1:4        200.0.0.1:4
---  201.0.0.2         192.168.30.10      ---                ---
```

d- Verify Connectivity from Outside the Private Network:

- **From a PC outside the private LAN** (for example, PC4).
- **Ping the public IP of the NAT interface** 172.40.0.5

```
C:\>ping 172.40.0.5

Pinging 172.40.0.5 with 32 bytes of data:

Reply from 130.0.0.1: Destination host unreachable.
Reply from 130.0.0.1: Destination host unreachable.
Request timed out.
Reply from 130.0.0.1: Destination host unreachable.

Ping statistics for 172.40.0.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

From PC4 (outside the private LAN), pinging the public IP 172.40.0.5 results in 100% loss. Thisis expected, as NAT only translates outgoing traffic from the private PCs (PC6-PC10). The private PCs cannot be reached directly from outside. From PC6-PC10, pinging any address in the external network (for example, Router0 or PC4 if routing allows) succeeds, showing that PATis working and all five PCs share the public IP correctly

Pinging on PC6 should succeed:

```
C:\>ping 130.0.0.10

Pinging 130.0.0.10 with 32 bytes of data:

Reply from 130.0.0.10: bytes=32 time=20ms TTL=126
Reply from 130.0.0.10: bytes=32 time=1ms TTL=126
Reply from 130.0.0.10: bytes=32 time=12ms TTL=126
Reply from 130.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 130.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 8ms
```

Numbered ACL on Router0 to deny PC5 from accessing VLAN 110

Deny PC5 (130.0.0.11) from accessing VLAN 110 (110.0.0.0/24):

```
Config terminal

access-list 110 deny ip host 130.0.0.11 110.0.0.0 0.0.0.255

access-list 110 permit ip any any

interface s0/1/0

ip access-group 110 in

exit
```

Verifying numbered ACL to deny PC5 from VLAN 110:

Trying to ping PC0 (110.0.0.1) from PC5 -> should fail

```
C:\>ping 110.0.0.1

Pinging 110.0.0.1 with 32 bytes of data:

Reply from 200.0.0.1: Destination host unreachable.
Reply from 200.0.0.1: Destination host unreachable.
Reply from 200.0.0.1: Destination host unreachable.
Reply from 200.0.0.1: Destination host unreachable.

Ping statistics for 110.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Trying to ping from PC4 → should succeed.

```
C:\>ping 110.0.0.1

Pinging 110.0.0.1 with 32 bytes of data:

Reply from 110.0.0.1: bytes=32 time=12ms TTL=254
Reply from 110.0.0.1: bytes=32 time=1ms TTL=254
Reply from 110.0.0.1: bytes=32 time=1ms TTL=254
Reply from 110.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 110.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 3ms
```

Named ACL "ALHITTP" to deny PC4 from HTTP/HTTPS from server:

On Router1

***Note: it was named ALHITTP (there is an "I" after the H)**

PC4 IP = 130.0.0.10

Server IP = 192.168.30.10

```
Config terminal

ip access-list extended ALHTTP

deny tcp host 130.0.0.10 host 192.168.30.10 eq 80

deny tcp host 130.0.0.10 host 192.168.30.10 eq 443

permit ip any any

exit

interface GigabitEthernet0/0/0.30

ip access-group ALHITTP in

exit
```

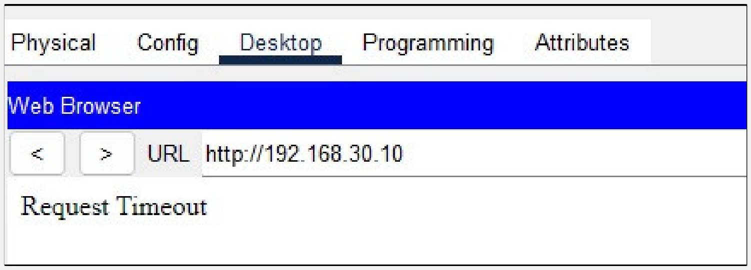
Verifying named ACL works correctly:

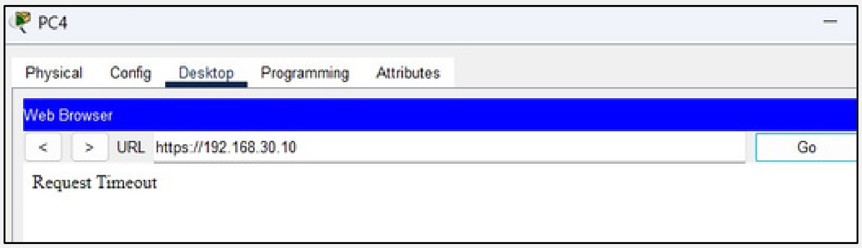
a- **Test from PC4:**
Try to ping the server (192.168.30.10) - ping should succeed.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 201.0.0.2: bytes=32 time=13ms TTL=126
Reply from 201.0.0.2: bytes=32 time=17ms TTL=126
Reply from 201.0.0.2: bytes=32 time=10ms TTL=126
Reply from 201.0.0.2: bytes=32 time=7ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 17ms, Average = 11ms
```

b- Check ACL hits on Router1:

Opening a web browser on PC4 and access HTTP (http://192.168.30.10) and HTTPS (https://192.168.30.10) - these should fail or time out.





Use show access-lists ALHITTP

```
    permit ip any any

R1-Psut#show access-list ALHITTP
Extended IP access list ALHITTP
    deny tcp host 130.0.0.10 host 192.168.30.10 eq www (12 match(es))
    deny tcp host 130.0.0.10 host 192.168.30.10 eq 443
    permit ip any any

R1-Psut#
```

c- Verify ACL applied on interface:

Using show running-config (on Router1) must display *ip access-group ALHITTP in*

```
interface GigabitEthernet0/0/0.130
 encapsulation dot1Q 130
 ip address 130.0.0.1 255.255.255.0
 ip access-group ALHITTP in
```

Conclusions

This project successfully demonstrates the integration of multiple core networking concepts into a functional network. By implementing VLAN segmentation, DHCP services, OSPF routing, NAT/PAT, and ACLs, the network achieves both connectivity and security goals. The simulation reinforces practical skills in Cisco Packet Tracer and mirrors scenarios encountered in professional network administration. This project highlights the importance of structured network design, adherence to security policies, and thorough configuration testing to ensure optimal performance.

References

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